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EVALUATION OF THE IMPACT OF CHANNEL DEFORMATIONS ON THE RELIABILITY OF OBSERVATION DATA AT THE QIPCHOQ GAUGING STATION

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Abstract — *This paper evaluates the impact of riverbed deformations on the reliability of hydrological observation data, using the Qipchoq gauging station in the Lower Amu Darya basin as a case study. Hydrological records spanning the last decade (2015–2025) and field survey data obtained via modern GNSS F4/F7 receivers and a SonTek M9 Acoustic Doppler Current Profiler (ADCP) were analyzed. The findings reveal a critical decrease in annual runoff volume by approximately 7 billion m³, alongside a 30–35% decline in interannual discharge trends. This has triggered intensive sedimentation within the riverbed (with silt accumulation thickness ranging from 75 cm to 2.41 m) and the formation of stable sand islands. Substantial discrepancies and inaccuracies between the existing rating curves (gauge height readings) and actual cross-sectional discharge measurements are substantiated. Based on these findings, the construction of a settling basin upstream of the gauging station is scientifically proposed to ensure precise water accounting.*

Keywords — *Lower Amu Darya, Qipchoq gauging station, channel deformation, water discharge, sediment accumulation, Doppler current profiler, GNSS measurements, rating curve.*

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1. INTRODUCTION

Water resources form the vital foundation for the socioeconomic development of modern society. Regulating river runoff using large-scale water intake systems represents one of the primary pathways in integrated water resources management [1]. Depending on the annual water abundance levels, intensive channel processes are observed across river systems. Full and intensive utilization of water resources has driven severe erosion, riverbed deformation, and the continuous accumulation of heavy sediment loads [2, 3]. Consequently, unfavorable hydrological conditions have emerged within river basins, compromising the operational reliability of water supply managed via reservoirs and downstream intake structures. In this regard, long-term monitoring and research conducted in the lower reaches and deltas of rivers provide an opportunity to explore key issues related to flow hydrology and channel dynamics [4, 5]. Special attention is currently directed toward the temporal management of river runoff volumes and the development of advanced methods for damless water diversion supported by channel regulation structures. The evolution of channel processes into an independent scientific discipline occurred during the 1940s and 1950s, marked by the publication of major foundational works by M.A. Velikanov, N.I. Makkaveev, E.V. Shantser, N.E. Kondratiev, I.V. Popov, and N.A. Rzhanitsyn [6, 7]. The formation of

this new field advanced alongside a deeper substantiation of its theoretical and physical principles [8, 9]. Modern frameworks have introduced new classifications of channel-forming processes, proposed defining criteria for specific channel regimes, and established indicators to categorize riverbed typologies. The interaction between riverbeds and stream flows has been reinterpreted, uncovering regularities in how structural configurations and morphological characteristics adjust in response to prevailing channel types [10]. The interaction of river flow with channel boundaries is evaluated by incorporating factors that dictate water discharge variation, sediment transport, and hydrological regime transformations [11, 12, 13]. In recent years, severe bank erosion and excessive sediment concentrations within incoming flows have created substantial operational challenges for hydraulic structures [14]. Investigating hydrological and channel dynamics through direct field surveys remains essential for addressing riverbed degradation, sedimentation management, and broader water resource challenges. The objective of this research is to identify the critical factors negatively affecting water accounting in the Lower Amu Darya region, evaluate their impact on the accuracy of discharge measurements at the Qipchoq gauging station, and scientifically substantiate effective remediation measures.

2. MATERIALS AND METHODS

The study analyzes scientific-practical literature and multi-year field observation datasets on riverbed dynamics. A comprehensive database tracking the flow volumes of the Lower Amu Darya from 2015 to 2025 was constructed using official reports from relevant state authorities and departments. Water discharge records from the Qipchoq gauging station over the past 10 years were critically reviewed. Consultation meetings were conducted with specialists from the departments of "Water Resources, Hydrometry, and Dispatch Services" and "Organization of Repair and Rehabilitation Works for Hydraulic Structures" to compile historical documentation on recent morphological shifts around the station. Field measurements were performed at the Qipchoq station cross-section to refine available hydrographic datasets. Parameters including water discharge, flow velocity, depth profiles, surface width, and morphological characteristics were measured, confirming severe riverbed deformation at the active monitoring cross-section. The field investigations adhered to standard hydraulic laws and hydrographic techniques. Benchmarks established along the riverbanks were used to determine reference elevations. To determine water surface elevations, a GNSS F4 receiver was set up over a known benchmark to serve as the reference base station operating across X, Y, and Z axes. Concurrently, a GNSS F7 receiver acted as the rover unit, calibrated to an accuracy range of 0.001–0.002 meters to pinpoint precise target elevation coordinates. Riverbed topography and depth profiles were mapped using a SONTEK M9 Acoustic Doppler Current Profiler (ADCP), capturing water depth measurements at discrete intervals of 2–3 cm across the entire channel cross-section.

3. RESULTS

Analysis of annual water discharge and volume data over the last decade at the Qipchoq gauging station reveals a highly dynamic flow variation, oscillating between 7 billion m^3 and 14.5 billion m^3 per year. The 10-year trend highlights a critical decline in the total annual runoff volume, which has decreased by nearly 7 billion m^3.

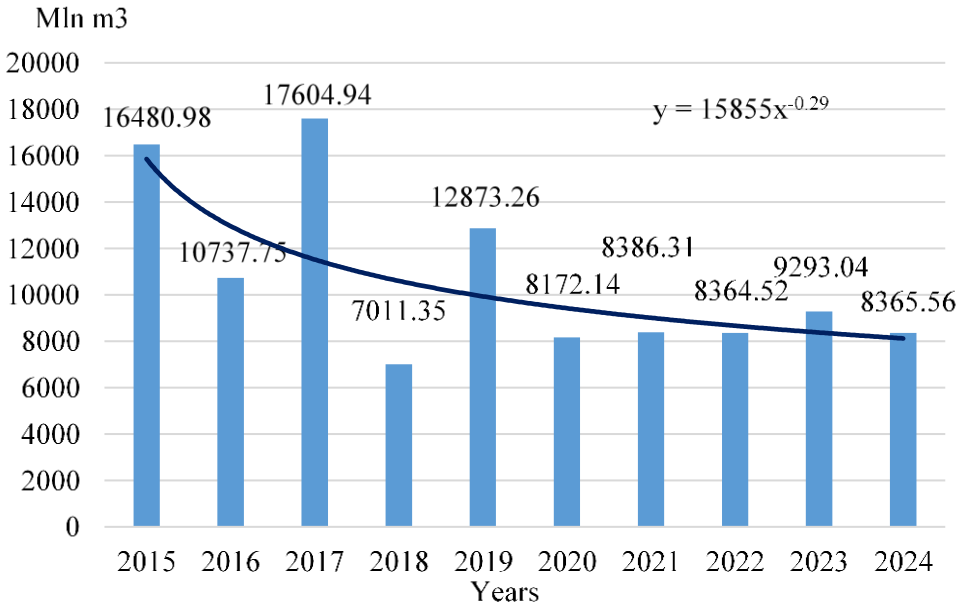


Figure 1. Flow volume at the Kipchak hydropost

Daily and monthly average water discharge rates over this 10-year period varied from 50 m³/s to 1700 m³/s. Peak discharge levels reaching 1500–1700 m³/s were observed in only two specific years (2015 and 2017). Throughout the remaining years, maximum discharge did not exceed 800 m³/s. This prolonged exposure to low-flow conditions led to accelerated deposition of suspended sediments within the channel, resulting in the formation of permanent sandbars and mid-channel islands. These changes severely obstructed the execution of reliable hydrometric measurements at the original cross-section, ultimately necessitating the physical relocation of the gauging station to a downstream reach.

The long-term discharge trend at the Qipchoq gauging station indicates a 30–35% reduction in available flow volumes. This decline progressively impairs the operational efficiency of downstream hydraulic infrastructure in the Lower Amu Darya and accelerates the development of stable island formations and wetland zones within the primary channel.

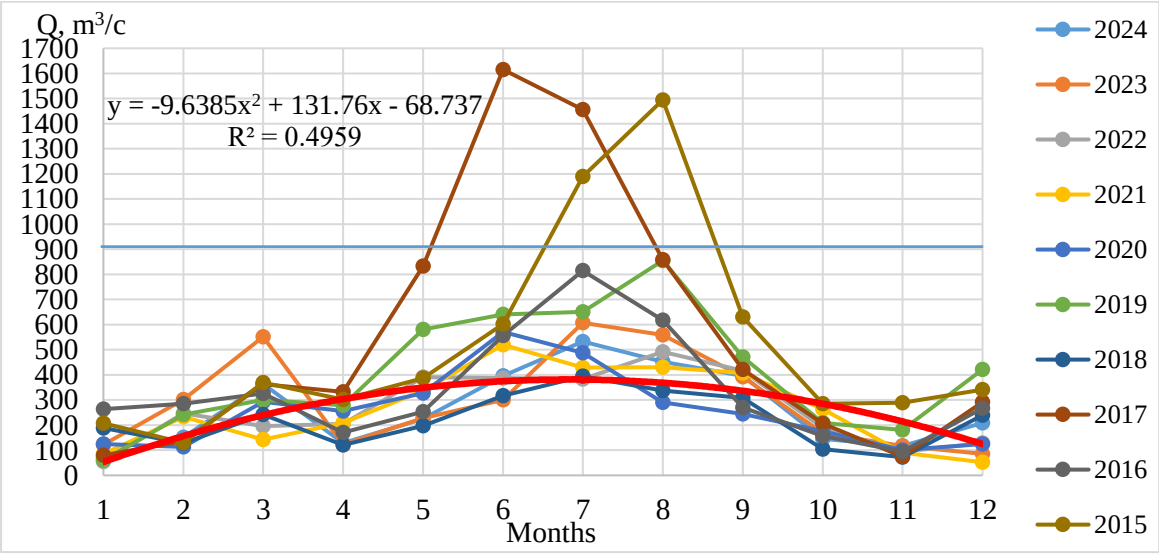


Figure 2. Monthly water discharge from the Kipchak hydropost

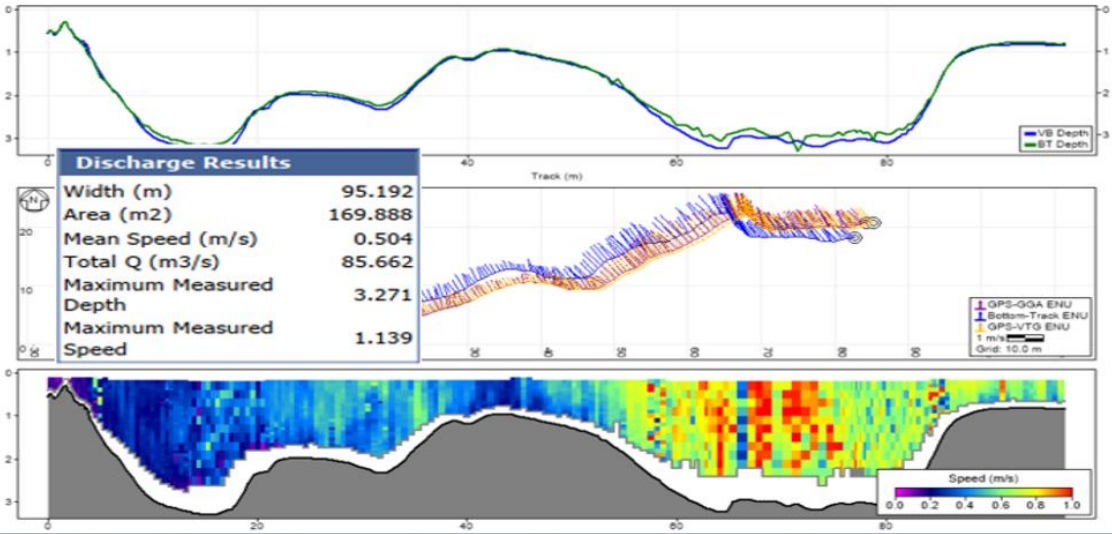
To evaluate the accurate water surface elevations at the station, existing benchmark references were audited against our field survey results. The comparative baseline data is structured in Table 1.

Table 1. Measured water surface elevations and benchmark references at the Qipchoq gauging station

Observation Points	Latitude (X)	Longitude (Y)	Elevation (Z), m
Base Benchmark Elevation	-	-	84.45
Water Surface Elevation (General)	-	-	80.12
Installed Base Benchmark Coordinate	4680718.4532	261474.3338	84.4540
Identified Water Level Elevation (Point 1)	4680709.2414	261517.0343	80.1180
Identified Water Level Elevation (Point 2)	4680933.1920	261547.8439	80.0957
Identified Water Level Elevation (Point 3)	4680960.7047	261758.0477	80.1073

The high-precision field survey identified an actual water surface elevation of 80.11 m. The absolute variance against the reference base benchmark was restricted to 0.01 m, yielding a negligible error margin of 0.01%, which confirms the high precision of the field data.

Direct field cross-sectional surveys executed on December 2, 2025, recorded the following morphological and hydraulic parameters: channel width of 95.19 m, cross-sectional area of 169.89 m², mean flow velocity of 0.50 m/s, maximum recorded flow velocity of 1.14 m/s, and a maximum channel depth of 3.27 m. The actual verified water discharge was established at 85.66 m³/s, confirming a corresponding water surface elevation of 80.120 m.



Gauge indicator	402	501	527	567	576	588	590	613
Measured	85.00	267.00	305.00	496.00	477.00	535.00	562.00	614.00
Graph	68.40	256.60	327.00	448.80	480.00	530.80	540.00	663.70
Variation	16.60	10.40	-22.00	47.20	-3.00	4.20	22.00	-49.70

Figure 3. Measurement results at the Kipchak hydropost

Comparing the station's historical rating curves (staff gauge height records) against the newly captured field survey datasets exposed distinct systematic discrepancies and non-conformities:

- At low-flow stages with discharges up to 250 m³/s, historical rating curves systematically overestimate the actual measured discharge rates.
- Within the mid-range interval of 250–350 m³/s, the rating curve values align tightly with field measurements.
- In the 350–550 m³/s discharge range, actual measured flows fall below the values predicted by historical rating curves.
- Conversely, when water discharge exceeds 550 m³/s, direct field measurements yield values higher than the historical rating curves.

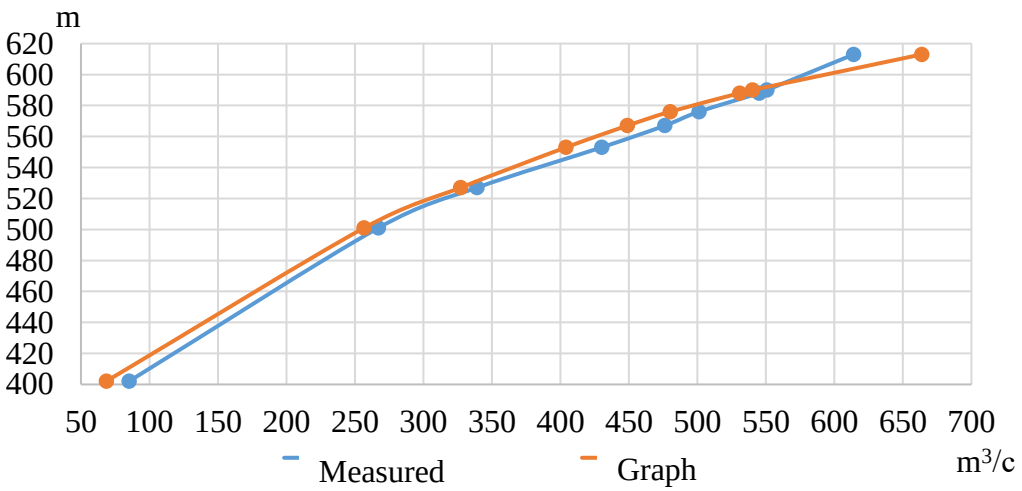


Figure 4. Measurement and graphical difference in water discharge at the hydropost

These shifts are driven by continuous channel bed deformations and structural modifications of the riverbed topography caused by localized sediment redistribution.

4. DISCUSSION AND CONCLUSION

Analysis of the bed topography reveals that for depth calculation, setting the deepest point of the channel (elevation 76.100 m) to a conditional "0" relative to the water surface elevation (80.120 m) corresponds to a staff gauge reading of 402 cm, representing an officially recorded discharge of 94.5 m³/s. However, direct ADCP measurements demonstrated an actual maximum depth of 3.27 m and an average depth of 1.60 m.

Subtracting the verified maximum depth (3.27 m) from the water surface elevation (80.120 m) isolates the true bed elevation at 76.85 m. This indicates that the riverbed at the Qipchoq gauging station cross-section has accumulated a heavy silt layer up to 75 cm thick. Incorporating average depth profiles shows that subtracting 1.61 m from the 80.120 m elevation reference isolates a bed level at 78.51 m, revealing that localized sediment deposition thickness reaches up to 2.41 m across certain sections. Based on the channel width and geometry of the sediment profile, the total volume of accumulated

sediment within a 1 km monitoring reach surrounding the gauging station is calculated at approximately 250,000 m^3.

To secure the long-term accuracy of hydrometric accounting and mitigate the impacts of channel deformation, it is recommended to construct a specialized sediment settling basin upstream of the Qipchoq gauging station cross-section and introduce a protocol for regular recalibration of the station's hydraulic rating curves.

REFERENCES

1. 1. Bakiev M.R., Tursunov T.N., Kaveshnikov N.T. Operation of Hydraulic Structures. Tashkent. 2008. (In Uzbek)

2. Valiev Sh.F., Usupaev Sh.E. Siltation as an indicator of management quality of regulated river runoff. Science and New Technologies, No. 6, 2013. Pp. 46-50. (In Russian)

3. Danelia N.F. Water Intake Structures on Rivers with Abundant Bed Load. Moscow. Kolos. 1964. P. 336. (In Russian)

4. Mukhammedov A.M. Operation of Low-Head Hydraulic Facilities on Sediment-Transporting Rivers (Case Study of Central Asia). Fan. Tashkent. 1976. P. 237. (In Russian)

5. Ismagilov Kh.A., Sayidov M.T. Riverbed processes in the backwater reduction zone of a reservoir. Hydrotechnical Construction Journal, No. 5. Moscow: 2014. Pp. 49-51. (In Russian)

6. Mikhailov V.N., Rogov M.M., Chistyakov A.A. River Deltas: Hydrological and Morphological Processes. Leningrad. Gidrometeoizdat, 1986. P. 280. (In Russian)

7. Rogov M.M. Hydrology of the Amu Darya Delta: Geographical and Hydrological Characteristics. Leningrad: Gidrometeoizdat, 1957. P. 257. (In Russian)

8. Baymanov K.I., Uzakov T.J., Baimanov R.K. and Tajibaev Sh. Riverbed processes in the lower reaches of the Amu Darya river in conditions of anthropogenic impact on the river flow. IOP Conference Series: Earth and Environmental Science. Pp. 1-9.

9. Mukhamedov A.M., Tuzov V.E. Features of channel deformations of the Amudarya River (dengish). Proceedings of coordination meetings on hydraulic engineering (Leningrad: Energy), 1967. 36. Pp. 244-258.

10. Rzhanitsyn N.A. Riverbed-Forming Processes of Rivers. Leningrad. Gidrometeoizdat, 1985. P. 263. (In Russian)

11. Velikanov M.A. Channel Process. Moscow, Fizmatgiz, 1958. P. 396. (In Russian)

12. Velikanov M.A. Dynamics of Alluvial Flows. Moscow, Fizmatgiz, 1955. P. 237. (In Russian)

13. Makkaveev N.I. Runoff and Channel Processes. Moscow, MGU., 1971. P. 115. (In Russian)

14. Makkaveev N.I. Erosion-Accumulation Processes and River Channel Topography: Selected Works. Moscow, MGU. 1998. P. 292. (In Russian)

15. Petrov A.A., Kabilov Kh.I., Sabirov M.R. and Jurayev B.B. Supplying long-term workability of the old elements of the hydraulic facilities by using of cold compositions. E3S Web of Conferences 392, 01038 (2023).

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SUV KAMI, O'RTACHA SUVLILIK VA SUV KO'P YILLAR UCHUN AMUDARYO SUV OMBORLARINING OPTIMAL ISH REJIMI

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Annotatsiya / Abstract — Ushbu maqolada Amudaryo havzasida suv resurslarining taqchilligi va iqlim o'zgarishlari sharoitida turli gidrologik suvlilik yillari uchun Nurek va Tuyamuyun suv omborlari majmuasining (TMGU) kafolatlangan suv sarfi hamda optimal ish rejimlari tahlil qilingan. Kam suvlilik yillarida Nurek suv omboridan chiqariladigan irrigatsion-ekologik popusklarni quyi oqim talabiga moslashtirish, TMGUning O'zan suv omboridan suvni chuqur va muhofazalangan Kaparas hamda Sultonsandjar suv omborlariga o'tkazish orqali bug'lanish yo'qotishlarini 15-25% ga kamaytirish muhandislik nuqtai nazaridan asoslangan. Ko'p suvlilik yillarida esa toshqin piklarini yumshatish, TMGU quyi b'yefida qirg'oq yemirilishi (degish) xavfining oldini olish hamda Orolbo'yi deltasiga kafolatlangan 10.0-15.0 km³/yil ekologik suv tashlamalarini yetkazib berish mexanizmi ko'rib chiqilgan.

Kalit so'zlar / Keywords — Amudaryo, Nurek suv ombori, Tuyamuyun suv omborlari majmuasi (TMGU), kafolatli suv sarfi, suv balansi, bug'lanish yo'qotishlari, ekologik suv tashlamasi, gidrodinamik modellashtirish.

1. KIRISH (INTRODUCTION)

Amudaryo — Markaziy Osiyodagi eng yirik, gidravlik jihatdan murakkab va strategik muhim transchegaraviy suv arteriyasi hisoblanadi [1, 2]. Havza hududida joylashgan mamlakatlar va mintaqalar o'rtasida suv resurslarini adolatli va oqilona boshqarish yuqori oqimdagi energetik va quyi oqimdagi irrigatsion-ekologik manfaatlar o'rtasidagi murakkab gidravlik muvozanatni talab etadi [3]. So'nggi yillarda olib borilgan ko'plab ilmiy tadqiqotlar ko'rsatishicha, global iqlim o'zgarishi, tog' tizmalaridagi muzliklar maydonining shiddat bilan qisqarishi hamda qurg'oqchilik davrlarining tezlashishi havza suv balansiga salbiy ta'sir ko'rsatmoqda [5]. Ayniqsa, kam suvlilik (75-85%) va qurg'oqchilik kuzatilgan yillarda Amudaryoning quyi oqimida suv tanqisligi darajasi 30-35% gacha yetib, qishloq xo'jaligi va aholini ichimlik suv bilan ta'minlashda jiddiy muammolarni keltirib chiqarmoqda [1, 4].

Ushbu murakkab suv-ekologik muammolarni hal etish, suv resurslarini kafolatli boshqarish va yo'qotishlarni kamaytirish maqsadida mahalliy va xorijiy olimlar tomonidan keng ko'lamli tadqiqotlar olib borilgan. Ilmiy manbalar va adabiyotlar tahlili quyidagi asosiy yo'nalishlarni qamrab oladi: Vaxsh daryosida joylashgan Nurek suv ombori va Amudaryoning quyi oqimidagi TMGU o'rtasidagi masofa 1000-1200 km ni tashkil etadi [8, 9]. O'zan suv омбори keng va kam chuqurlikka ega bo'lib, yoz oylarida bug'lanish 1200-1400 mm ga yetadi [2, 4, 10].

2. MATERIAL VA USULLAR (MATERIALS AND METHODS)

Tadqiqotni o'tkazishda foydalanilgan metodologiya, eksperimental qurilmalar va tahlil usullari yoritiladi. Suv omborlarining suv balansi tenglamasi quyidagicha keltiriladi:

$$W_{end} = W_{start} + Q_{in} \cdot \Delta t - Q_{out} \cdot \Delta t - Q_{lost} \cdot \Delta t \qquad (1)$$

Bu yerda: W_{start} — mavsum boshida hajm, mln m³; Q_{in} — kiruvchi suv sarfi, m³/s; Q_{out} — chiquvchi suv sarfi, m³/s; Qlost — suv yo'qotilishi (filtratsiya va bug'lanish), m³/s; Δt — ko'rilayotgan vaqt oralig'i. Formulalar MS Equation yoki MathType dasturi orqali kiritilib, sahifaning o'ng cheti bo'ylab raqamlanadi.

3. NATIJALAR (RESULTS)

Olingan ilmiy va amaliy natijalar xolis holda taqdim etiladi. Maqoladagi jadvallar va rasmlar tartib bilan raqamlangan hamda sarlavhaga ega bo'lishi shart.

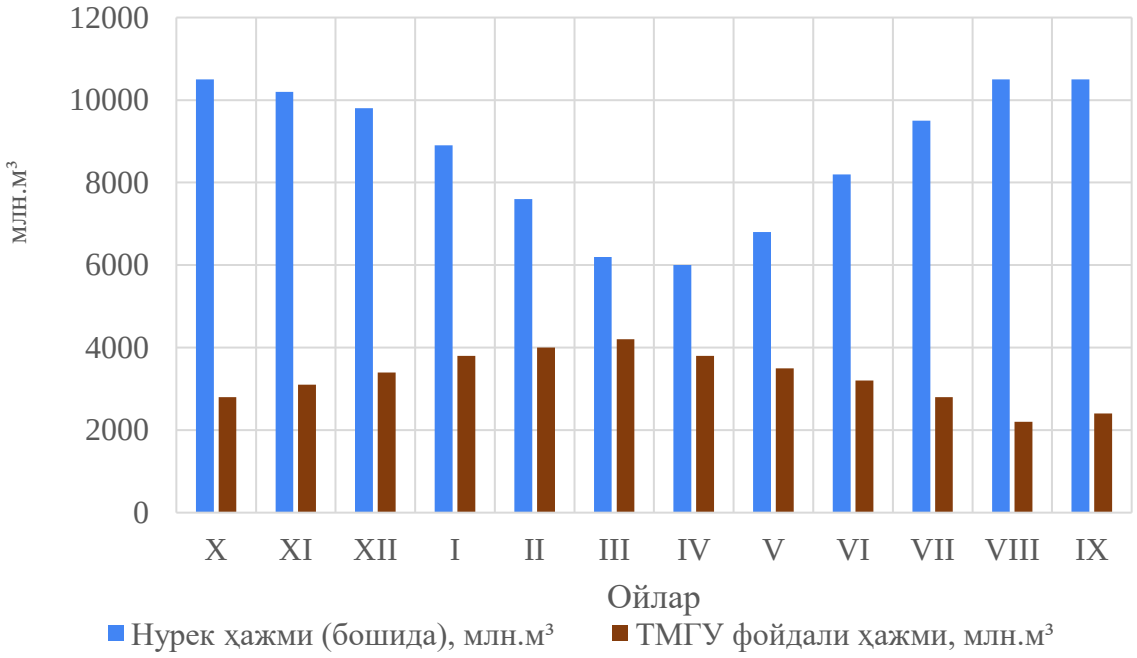
1-JADVAL

Vegetatsiya davrida daryo oqimining boshqarilishi (Kam suvlilik yili P=85%)

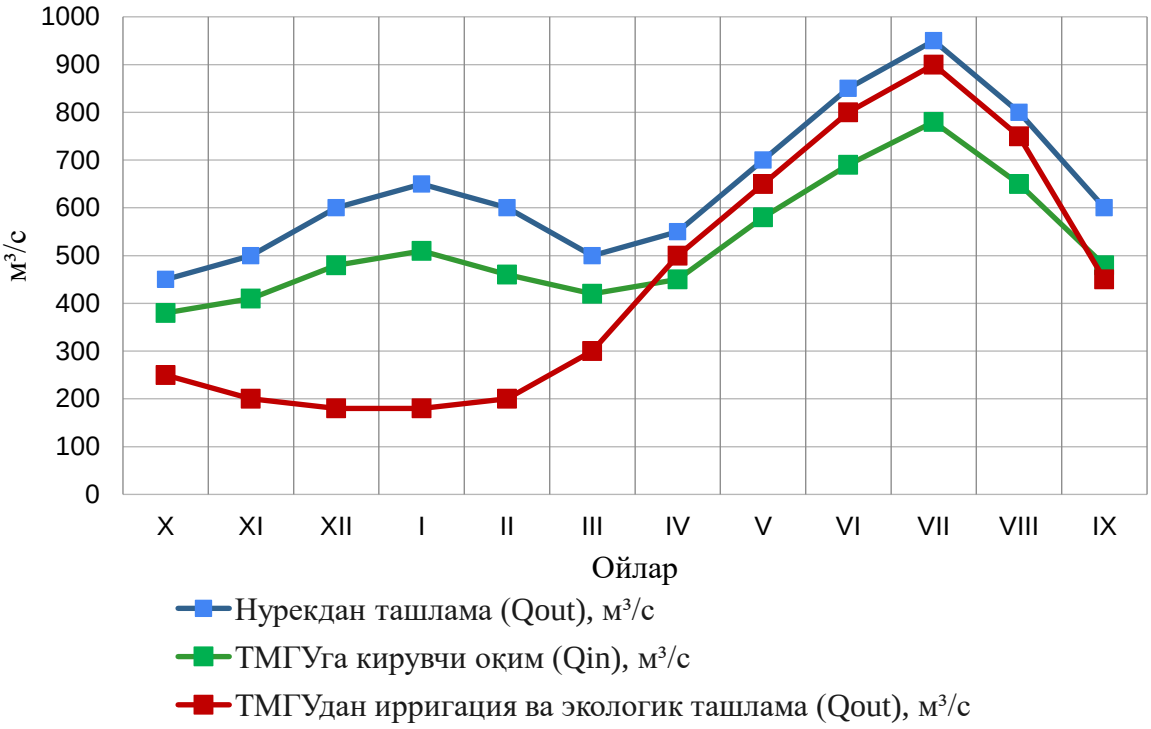
Oylar	Nurek hajmi mln.m³	Nurekdan tashlama (Q_{out}), m³/s	TMGUga kiruvchi oqim (Q_{in}), m³/s	TMGU foydali hajmi, mln.m³	TMGUdan tashlama (Q_{out}), m³/s	Dispetcherlik operativ rejim
X	10,500	450	380	2,800	250	Novegetatsiya boshlanishi, Nurekda maksimal hajm
XI	10,200	500	410	3,100	200	Energetik rejim, TMGUda suv to'plash
XII	9,800	600	480	3,400	180	Quyilma suv

Oylar	Nurek hajmi mln.m^3	Nurekdan tashlama (Q_{out}), m^3/s	TMGUga kiruvchi oqim (Q_{in}), m^3/s	TMGU foydali hajmi, mln.m^3	TMGUdan tashlama (Q_{out}), m^3/s	Dispetcherlik operativ rejim
						omborlari - Kaparasni to'ldirish
I	8,900	650	510	3,800	180	Qishki energetik rejim
II	7,600	600	460	4,000	200	TMGUda maksimal zaxira yaratish
III	6,200	500	420	4,200	300	Sho'r yuvish davri boshlanishi
IV	6,000	550	450	3,800	500	Vegetatsiya boshlanishi, ekinlarni sug'orish
V	6,800	700	580	3,500	650	Nurek suv to'plashni boshlaydi, TMGUdan suv berish
VI	8,200	850	690	3,200	800	Pik irrigatsiya davri, suv

Oylar	Nurek hajmi mln.m^3	Nurekdan tashlama (Q_{out}), m^3/s	TMGUga kiruvchi oqim (Q_{in}), m^3/s	TMGU foydali hajmi, mln.m^3	TMGUdan tashlama (Q_{out}), m^3/s	Dispetcherlik operativ rejim
						tanqisligini yopish
VII	9,500	950	780	2,800	900	Nurekdan maksimal tashlama, Kaparas, Sultonsandjar, Qo'shbuloqdan foydalanish
VIII	10,500	800	650	2,200	750	Vegetatsiya yakunlanishi, tejamkorlik rejimi
IX	10,500	600	480	2,400	450	Vegetatsiya oxiri, keyingi mavsumga tayyorgarlik



Nurek suv ombori (Vaxsh) va Tuyamuyun gidrouzeli (Amudaryo quyi oqimi) foydali suv hajmining (W, mln m³) mavsumiy va yillik o'zgarish dinamikasi



Suv tashlamalari va suv sarfi dinamikasi (m³/s)

2-JADVAL

O'rtacha suvlilik (P=50%) va ko'p suvlilik (P=25%) yillari uchun asosiy rejim ko'rsatkichlari

Gidrologik sharoit	Nurek suv ombori maks. hajmi (mln.m³)	Nurekdan maks. tashlama (m³/s)	TMGUga keluvchi oqim (m³/s)	TMGUdan tashlama (m³/s)	Asosiy muhandislik tadbiri
O'rtacha suvlilik (P=50%)	10,500	1,150	960	1,100	Kaparas va Sultonsandjarga suv o'tkazib bug'lanishni 15-20% kamaytirish
Ko'p suvlilik (P=25%)	10,500	1,450	1,220	1,400	Toshqin piklarini yumshatish, qirg'oq yemirilishini oldini olish, deltaga suv berish

[DIAGRAMMA VIZUALIZATSIYA: 3-RASM]

Ko'p suvlilik (P=25%) va o'rtacha suvlilik (50%) sharoitida suv sarflari dinamikasi diagrammasi

4. MUNOZARA VA XULOSA (DISCUSSION AND CONCLUSION)

Olingan natijalar boshqa mualliflarning ishlari bilan solishtiriladi va muhokama qilinadi. Yakunda tadqiqot bo'yicha umumiy xulosalar hamda kelgusidagi izlanishlar uchun tavsiyalar beriladi.

Xulosa: TMGUda suvni O'zan suv omboridan Kaparas va Sultonsandjarga operativ suv

o'tkazish orqali bug'lanish yo'qotishlarini 15-25% ga (yillik 180-220 mln m^3) qisqartirish imkoni mavjud. Quyi oqimning suv kafolatini oshirish uchun kam suvlilik yillarida Nurek va TMGU o'rtasida muvofiqlashtirilgan operativ dispetcherlik grafigini qo'llash Xorazm va Qoraqalpog'istondagi suv tanqisligi darajasini 35% dan 12% gacha pasaytirishi mumkin. Ko'p suvlilik yillarida Nurek va TMGU sig'imlari yordamida toshqin piklari 20-25% ga yumshatilib, Amudaryo quyi oqimidagi o'zan jarayonlari va qirg'oq yemirilishi xavfi keskin pasayadi.

ADABIYOTLAR RO'YXATI / REFERENCES

1. Ikramova M.R. va boshqalar. Amudaryo havzasida suv resurslarini optimal boshqarish va suv omborlari ish rejimlari tahlili. — Toshkent, 2024.
2. Mirzayev N.N. Irrigatsiya tizimlarida kafolatlangan suv ta'minoti va suv balansi hisoblari. — Toshkent: Fan, 2021.
3. Rahimov Sh.H., Jo'rayev A.Q. Daryo o'zanlarida gidrodinamik va o'zan jarayonlarini modellashtirish. — Toshkent: TIQXMMI, 2020.
4. Bakiyev M.R., Ibragimov I.A. Gidrotexnika inshootlari va ularning xavfsizligi. — Toshkent: TIQXMMI, 2019.
5. IPCC. Climate Change and Land: Water-Energy-Food Nexus in Arid Regions. — Cambridge University Press, 2022.
6. US Army Corps of Engineers. HEC-RAS River Analysis System: Hydraulic Reference Manual. — Hydrologic Engineering Center, 2023.
7. DHI. MIKE Hydro River and MIKE 21 Modelling Systems User Guide. — Danish Hydraulic Institute, 2022.
8. Baklanova D.V. Optimizatsiya rezjimov raboty kaskada vodohranilish v basseynah transgranichnyh rek. — Moskva: Kolos, 2018.
9. Duhovnyy V.A., Sokolov V.I. Integrirovannoe upravlenie vodnymi resursami v basseynye Aralskogo morya. — Tashkent: NIC MKVK, 2017.
10. Chechasov V.I., Sherfedinov L.Z. Regulirovanie stoka i isparenie s poverhnosti vodohranilish Sredney Azii. — Tashkent: Gidrometeoizdat, 2015.
11. Mavlonov A.A., Ergashev R.R. Ruslovye protsessy i yavleniya degisha v nizhnem techenii Amudari. — Toshkent: Fan, 2019.
12. Deltares. Delft3D-FLOW: Simulation of Multi-Dimensional Hydrodynamics and Sediment Transport. — Delft, Netherlands, 2021.
13. ESA. Sentinel-3 and Landsat Operational Satellite Monitoring for Inland Water Resources. — European Space Agency Technical Report, 2023.
14. Saliyev B.K., Mahmudov E.Zh. Suv resurslaridan oqilona foydalanish va avtomatik boshqarish tizimlari. — Toshkent: Zamonaviy fan, 2022.
15. MKVK. Mezhgosudarstvennoe soglasovanie limitov vodozaborov i rezhimov raboty vodohranilish v basseynye reki Amudarya. — Informatsionnyy byulleten NIC MKVK, 2023.